

Lecture 2

Row Reduction Algorithm

$$\begin{cases} x_1 + 2x_2 = -1 \\ 2x_1 + x_2 + x_3 = 1 \\ -x_1 + x_2 = -1 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 2 & 1 & 1 & 1 \\ -1 & 1 & 0 & -1 \end{array} \right] \begin{array}{l} \xrightarrow{-2x} \\ \xrightarrow{+1x} \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & -3 & 1 & 3 \\ 0 & 3 & 0 & -2 \end{array} \right] \xrightarrow{+1x} \left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & -3 & 1 & 3 \\ 0 & 0 & 1 & 1 \end{array} \right] \sim \times (-\frac{1}{3})$$

3 pivots corresponding to the unknowns. $\Rightarrow$ No free variables
so 1 solution.

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & -1/3 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right] \xrightarrow{-2x} \sim \left[\begin{array}{ccc|c} 1 & 0 & 2/3 & 1 \\ 0 & 1 & -1/3 & -1 \\ 0 & 0 & 1 & 1 \end{array} \right] \begin{array}{l} \xrightarrow{+1/3 x} \\ \xrightarrow{-2/3 x} \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1/3 \\ 0 & 1 & 0 & -2/3 \\ 0 & 0 & 1 & 1 \end{array} \right] \Rightarrow \left. \begin{array}{l} x_1 = 1/3 \\ x_2 = -2/3 \\ x_3 = 1 \end{array} \right\} \leftarrow \begin{array}{l} \text{solution} \\ \text{parametric desc.} \end{array}$$

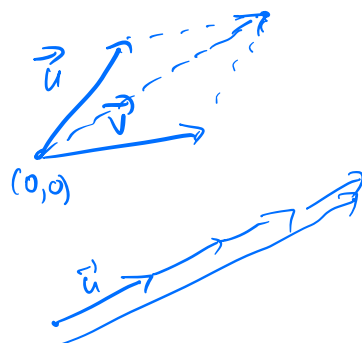
Important fact 1 | Row Reduction Algorithm: Matrix Multiplication

Example

$$\vec{u} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

addition: $\vec{u} + \vec{v} = \begin{pmatrix} 1+1 \\ -3+(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$

scalar multiplication: $4 \cdot \vec{u} = \begin{pmatrix} 4 \cdot 1 \\ 4 \cdot (-3) \end{pmatrix} = \begin{pmatrix} 4 \\ -12 \end{pmatrix}$



$$A = \begin{pmatrix} -2 & 3 & 1 \\ 5 & 0 & -4 \end{pmatrix} \quad (2 \times 3 \text{ matrix})$$

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Addition

$$A+B = \begin{pmatrix} -2+1 & 3+1 & 1+1 \\ 5+1 & 0+1 & -4+1 \end{pmatrix} = \begin{pmatrix} -1 & 4 & 2 \\ 6 & 1 & -3 \end{pmatrix}$$

Matrix-Vector multiplication (dot-product)

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad m \times n$$

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$A \cdot \vec{x} = \begin{bmatrix} a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n \\ \vdots \\ a_{m1} \cdot x_1 + a_{m2} \cdot x_2 + \dots + a_{mn} \cdot x_n \end{bmatrix}$$

columns of A
=
components of $\vec{x}$

Matrix Multiplication

$$(k \times m) \ A, \quad (m \times n) \ B$$

$$(\vec{b}_1 \parallel \vec{b}_2 \dots \vec{b}_n)$$

Columns of B

$$A \cdot B = (A \cdot \vec{b}_1 \quad A \cdot \vec{b}_2 \dots A \cdot \vec{b}_n)$$

$$\begin{pmatrix} 2 & 5 & -1 \\ 7 & -4 & 3 \end{pmatrix}^{2 \times 3} \cdot \begin{pmatrix} 4 & -1 \\ 2 & 3 \\ 0 & 1 \end{pmatrix}^{3 \times 2} = \left(\begin{pmatrix} 2 & 5 & -1 \\ 7 & -4 & 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 2 & 5 & -1 \\ 7 & -4 & 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} \right)$$

$$= \begin{pmatrix} 18 & 12 \\ 20 & -16 \end{pmatrix}$$

$$A \cdot B \neq B \cdot A$$

↑
in general

$$(A \cdot B)_{ij} = (a_{i1} \ a_{i2} \ \dots \ a_{im}) \cdot \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{mj} \end{pmatrix} = a_{i1} \cdot b_{1j} + \dots + a_{im} \cdot b_{mj}$$

↑
element of $A \cdot B$ on
 i 'th row & j 'th column

$$A^0 = I \leftarrow \text{identity matrix}$$

$$A^r \cdot A^s = A^{r+s} \leftarrow \text{for square matrices } (n \times n)$$

$$I_{n \times n} = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ & 0 & 1 & \\ \vdots & & & \ddots \\ & & & 0 & 1 \end{bmatrix}_n$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_m \cdot A = A = A \cdot I_n$$

$\downarrow$
 $m \times n$

$$A\vec{x} = \vec{b}$$

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 1 & 1 & 1 \\ -1 & 1 & 0 & -1 \end{bmatrix} \xrightarrow{-2x} \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & -3 & 1 & 3 \\ -1 & 1 & 0 & -1 \end{bmatrix} = A_2$$

$$A_2 = T \cdot A_1$$

$\downarrow$
?

$$T = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\leftarrow I_3$ with
the same
elementary row
operation applied
to it

$$\text{if } A \cdot B = 0 \not\Rightarrow A = 0 \text{ or } B = 0$$

$$\text{if } A \cdot C = A \cdot D \not\Rightarrow A = 0 \text{ or } C = D$$

$$A\vec{x} = \vec{b}$$

$$A: n \times n$$

A is regular (aka invertible) if $A \cdot C = C \cdot A = I$
for some $C, n \times n$

Singular otherwise

If A regular, $n \times n$

inverse A^{-1} : $A \cdot A^{-1} = A^{-1} \cdot A = I$

Important fact
2

A regular then $A\vec{x} = \vec{b}$ has the
unique solution $\vec{x} = A^{-1} \vec{b}$ for any
 $\vec{b} \in \mathbb{R}^n$

Example:

$$\begin{cases} -7x_1 + 3x_2 = 2 \\ 5x_1 - 2x_2 = 1 \end{cases}$$

$$A\vec{x} = \vec{b}$$

$$A = \begin{pmatrix} -7 & 3 \\ 5 & -2 \end{pmatrix} \quad \vec{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

A regular?

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

if $ad - bc \neq 0$ then
B is regular and

$$B^{-1} = \frac{1}{ad - bc} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

(only for 2×2)

$$(-7) \cdot (-2) - 3 \cdot 5 = 14 - 15 = -1$$

$$A^{-1} = \frac{1}{-1} \cdot \begin{pmatrix} -2 & -3 \\ -5 & -7 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

Check: $A \cdot A^{-1} = A^{-1} \cdot A = I_2$

Unique solution to $A\vec{x} = \vec{b}$ is $\vec{x} = A^{-1} \cdot \vec{b} = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
 $= \begin{pmatrix} 7 \\ 17 \end{pmatrix}$

Algorithm to find A^{-1} (if it exists)

1. Augmented Matrix $(A | I)$
2. reduced echelon form $(B | C)$
3. If $B = I$ then $A^{-1} = C$
 else, A is singular.

Example

1) $A = \begin{bmatrix} 2 & 1 & -4 \\ -4 & -1 & 6 \\ -2 & 2 & -2 \end{bmatrix}$ / $(A | I) = \left[\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ -4 & -1 & 6 & 0 & 1 & 0 \\ -2 & 2 & -2 & 0 & 0 & 1 \end{array} \right] \begin{matrix} \xrightarrow{+2} \\ \sim \\ \xrightarrow{+k} \end{matrix}$

$\sim \left[\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 2 & 1 & 0 \\ 0 & 3 & -6 & 1 & 0 & 1 \end{array} \right] \xrightarrow{-3r} \sim$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & -4 & 1 & 0 & 0 \\ 0 & 1 & -2 & 2 & 1 & 0 \\ 0 & 0 & 0 & -5 & -3 & 1 \end{array} \right]$$

↑ This can never give I_3
 so A is singular.

Note: In the lecture we unintentionally worked with the same matrix, but with a "+1" in row 2, column 2, instead of a "-1". The resulting matrix is regular, but the calculations become less nice.

$$2) A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$(A | I) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{-3 \times} \sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right] \xrightarrow{\times(-\frac{1}{2})}$$

$$\sim \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3/2 & -1/2 \end{array} \right) \xrightarrow{-2 \times} \sim \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 3/2 & -1/2 \end{array} \right]$$

$$A \text{ is invertible \& } A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$

Transpose of a matrix

$A: m \times n$

A^T : transpose of A

$$(A^T)_{ij} = A_{ji}$$

Example

$$A = \begin{pmatrix} -5 & 2 \\ 1 & -3 \\ 0 & 4 \end{pmatrix} \quad A^T = \begin{pmatrix} -5 & 1 & 0 \\ 2 & -3 & 4 \end{pmatrix}$$

Properties of Inverse/Transpose

$$\bullet (A^T)^T = A$$

$$\bullet (A \cdot B)^T = B^T \cdot A^T$$

$$\bullet (A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$$

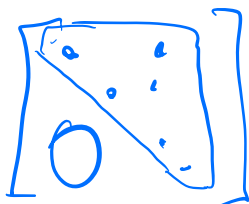
$$\bullet (A^T)^{-1} = (A^{-1})^T$$

↑
regular

more properties in reader

upper triangular

$n \times n$



lower triangular

$n \times n$



$$A \sim B \sim C = \dots \sim I$$

$\uparrow$
 reduced echelon
 form

$$B = T_1 \cdot A$$

$$C = T_2 \cdot B$$

$$I = \underbrace{T_k \cdot T_{k-1} \cdot T_{k-2} \cdot \dots}_{A^{-1}} \cdot A$$